Avatar Runtime

Purpose

Avatar Runtime is responsible for the physical embodiment of the digital character.

Its purpose is not rendering.

Its purpose is making the character visually believable.

Everything the user can physically observe about Shio belongs to Avatar Runtime.

The Runtime translates the current Character State into visible behavior while remaining completely independent from conversation logic, reasoning and external platforms.

Responsibilities

Avatar Runtime owns every aspect of the character's physical presence.

Its responsibilities include:

- VRM loading;
- animation playback;
- expression control;
- procedural blinking;
- procedural breathing;
- idle micro-movements;
- lipsync;
- gaze control;
- body posture;
- animation blending;
- visual state transitions.

Avatar Runtime is the body of the character.

It never decides what the character should do.

It only decides how the character appears.

Non-Responsibilities

Avatar Runtime deliberately does **not** own:

- dialogue;

- reasoning;
- memory;
- personality;
- speech recognition;
- tool execution;
- external integrations;
- moderation;
- streaming platforms.

These responsibilities belong to other Runtime systems.

Maintaining these boundaries keeps the visual system independent from conversational logic.

The Body, Not The Brain

Avatar Runtime should be viewed as the body of Shio.

It does not think.

It does not remember.

It does not make decisions.

It reacts to the Character State in the same way a body reacts to a mind.

This distinction is one of the core architectural principles of Shio Runtime.

Rendering Is an Implementation Detail

The architecture intentionally avoids depending on any rendering technology.

Today Avatar Runtime may use:

- Three.js;
- VRM;
- VRMA animations.

Tomorrow it may use completely different technologies.

The responsibility of Avatar Runtime remains unchanged.

Its purpose is representing the physical state of the character.

Observable Layers

Avatar Runtime is composed of several independent visual layers.

Examples include:

Body Motion

Responsible for:

- locomotion;
- idle animations;
- gestures;
- posture.

Facial Expression

Responsible for:

- emotions;
- smiles;
- eye expressions;
- brows;
- mouth expressions.

Procedural Motion

Responsible for:

- blinking;
- breathing;
- idle sway;
- subtle head motion;
- micro-adjustments.

These movements exist continuously, even during silence.

They significantly strengthen the illusion of life.

Lipsync

Responsible for synchronizing speech with visible mouth movement.

Avatar Runtime owns visual lipsync.

Speech Runtime owns spoken audio.

This separation keeps audio generation and rendering independent.

Gaze

Responsible for visual attention.

Examples include:

- looking at the user;
- looking toward the active speaker;
- looking away while thinking;
- returning to neutral.

Gaze should reflect Character State rather than independent logic.

Character State Integration

Avatar Runtime continuously observes Character State.

Examples include:

```text id="avr01" Mode = Thinking

↓

Thinking Animation
```

```text id="avr02"
Emotion = Happy

↓

Happy Expression
```

```text id="avr03" Speech = Speaking

↓

Lipsync Enabled ```

Avatar Runtime never receives direct commands from other Runtime systems.

It reacts to observable state.
```

Animation Philosophy

Animations should communicate intent rather than simply produce motion.

Every movement should have meaning.

Examples:

A blink is not an animation.

It communicates life.

Breathing is not a procedural effect.

It communicates presence.

A gesture is not movement.

It communicates intention.

Avatar Runtime therefore prioritizes expressive clarity over animation complexity.

Layer Independence

Visual systems should remain independent.

For example:

Body animation should not directly control facial expression.

Breathing should not modify lipsync.

Blinking should not interrupt gestures.

Each subsystem contributes independently to the final presentation.

The visible character emerges from composition rather than monolithic animation logic.

Future Evolution

Avatar Runtime is expected to become increasingly sophisticated while preserving its architectural responsibility.

Potential future improvements include:

- procedural gestures;
- adaptive idle behavior;
- richer breathing;
- contextual gaze;
- emotional body language;
- layered animation scheduling;
- expressive timing.

Regardless of future capabilities, Avatar Runtime remains responsible for one thing only:

Representing the physical presence of the digital character.